

Lab_Syn_4.0

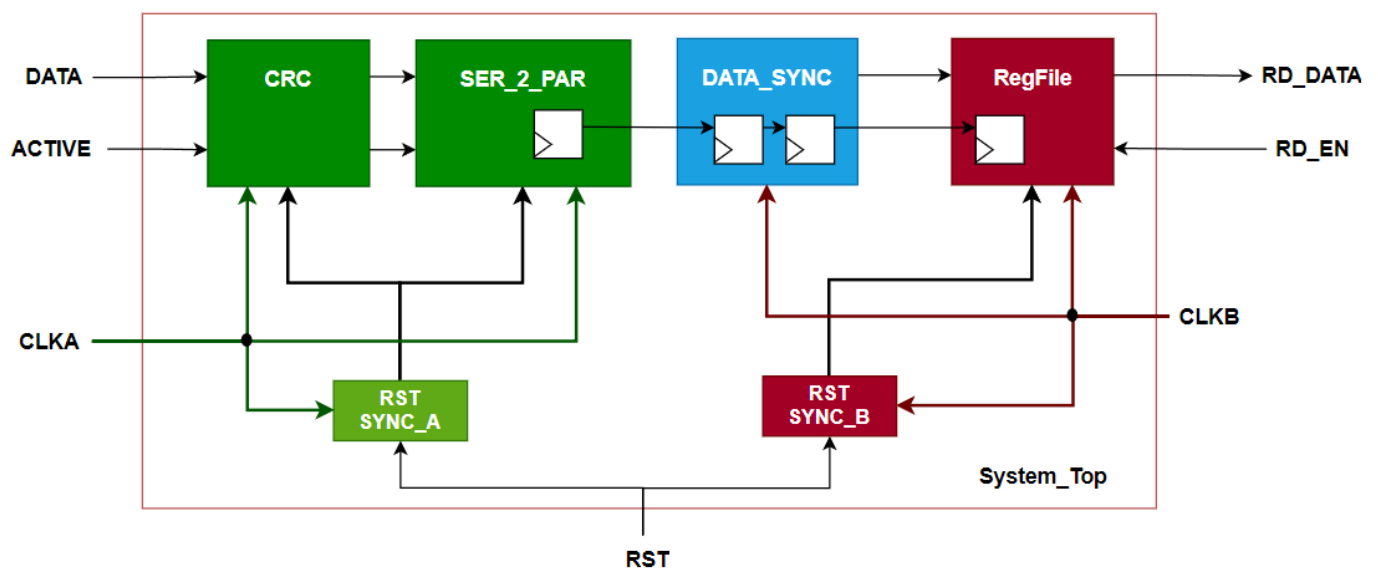
You have to synthesize **System_Top** hierarchy files using **Design Compiler** shell and generate **Technology Dependent** gate level netlists in Verilog format using standard cell libraries (typical, worst, best) inside **std_cells** folder:-

- scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db
- scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c
- scmetro_tsmc_cl013g_rvt_ss_1p08v_125c

Steps: -

1. Copy **Lab_Syn_4.0** folder from your computer into the virtual machine inside **Labs** directory.

Here is a simple block diagram for **System_Top** which represent the two asynchronous clocks **CLKA** and **CLKB** and their connections to help you to constraint our system



2. Open a terminal inside path **/home/IC/Labs/Lab_Syn_4.0/syn** and run **syn_script.tcl** script through design compiler shell using the following command in the terminal (**dc_shell -f syn_script.tcl | tee syn.log**)
3. Check the timing paths between **CLKA** and **CLKB**

4. Open **cons.tcl** file and Add the following constraints to exclude the timing paths between CLKA domain and CLKB domain: -

- **set_false_path** commands between CLKA to CLKB

OR

- **set_clock_groups** command between CLKA to CLKB

5. Open a terminal inside path **/home/IC/Labs/Lab_Syn_4.0/syn** and run again **syn_script.tcl** script through design compiler shell using the following command in the terminal (**dc_shell -f syn_script.tcl | tee syn.log**)

6. Check the timing paths between CLKA and CLKB again. It must be excluded and not included in the timing reports